2025 TRIAL

SEQUOIA — EDUCATION —

Specialist Mathematics Examination 2

Solutions Book

VCE Trial Examination – 2025

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Section A – Multiple-choice questions

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Question 1 (D)

The variable `count` tracks the number of times the **`while`** loop runs before termination. For each loop, if `n` is even, then it is halved and if `n` is odd, then it is multiplied by three and added to one. That is we get the following sequence for `n`:

`count`	0	1	2	3	4	5	6	7	8
`n`	6	3	10	5	16	8	4	2	1

When $n = 1$, the loop condition is no longer satisfied and the program terminates. Therefore, the final value of `count` is 8.

Question 2 (C)

The statement has the form $\forall m \in B,\ R(m)$, where B is the set of musicians in the band and $R(m)$ is the predicate 'm can read sheet music'. The negation of this statement is $\exists m \in B,\ \sim R(m)$, which reads 'there exists a musician in the band who cannot read sheet music', or more naturally, 'some musicians in the band cannot read sheet music'.

Question 3 (B)

We require that $-1 \le 2 - |x+1| \le 1 \implies -4 \le x \le -2$ or $0 \le x \le 2$. Hence, the maximal domain is $[-4,-2] \cup [0,2]$.

Question 4 (A)

Putting the equation in standard form, we get

$$\frac{(y-1)^2}{1^2} - \frac{(x-0)^2}{\left(\frac{1}{2}\right)^2} = 1.$$

The asymptotes pass through $(0,1)$ with gradients $\pm\dfrac{1}{\frac{1}{2}} = 2$. That is, the asymptotes are $y = 2x+1$ and $y = -2x+1$.

Question 5 (D)

For a point not on the real line, conjugation corresponds to a reflection in the real axis, and taking its negative corresponds to reflecting it in both the real and imaginary axes. These transformations are equivalent to a reflection in the imaginary axis.

Question 6 (B)

By De Moivre's theorem, an argument (modulo 2π) of $\dfrac{z^3}{w^2}$ is given by

$$\arg\left(\frac{z^3}{w^2}\right) = 3\operatorname{Arg}(z) - 2\operatorname{Arg}(w) = \frac{7\pi}{6}.$$

Therefore, the principal argument is

$$\operatorname{Arg}\left(\frac{z^3}{w^2}\right) = -\frac{5\pi}{6}.$$

Question 7 (D)

By the chain rule, we have

$$\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} = \frac{-\csc(t)\cot(t)}{-2\csc^2(t)} = \frac{1}{2}\cos(t).$$

Question 8 (A)

The suitable substitution is $u = 3 - x$ which gives

$$\frac{dx}{du} = -1, \quad u(0) = 3, \quad u(1) = 2, \quad x = 3 - u.$$

Therefore, we get

$$\int_0^1 \frac{x}{\sqrt{3-x}}\,dx = -\int_3^2 \frac{3-u}{\sqrt{u}}\,du = \int_2^3 \left(\frac{3}{\sqrt{u}} - \sqrt{u}\right) du.$$

Question 9 (B)

Option B is a well-known theorem, so is the correct answer. The other options are common misconceptions. **Option A** is wrong as demonstrated by the counterexample $f(x) = x + e^x$. It is strictly convex yet does not attain a global minimum. If a strictly convex function has a local minimum point, *then* one could conclude it is global. **Option C** is wrong since $f(x) = x^4$ has $f''(0) = 0$ despite being strictly convex. **Option D** is wrong since kf is strictly *concave* for any $k < 0$, not strictly convex.

Question 10 (D)

Along the x-axis the gradient of a solution curve is undefined, eliminating **option A** and **option C**. Where $x = 0$ and $y > 0$ the gradient of a solution curve is positive, so the answer is **option D**.

Question 11 (A)

Using $h = -0.25$, we get

$$y_1 = 1 - 0.25\left(0^2 + 1^2\right) = 0.75 \implies y_2 = 0.75 - 0.25\left((-0.25)^2 + 0.75^2\right) = 0.59375.$$

Question 12 (B)

We wish to find the time when the change in displacement of the particle is zero. That is, for $T \in [3,6]$,

$$0 = \int_0^T v(t)\,dt = \int_0^2 dt + \int_2^T (3-t)\,dt \implies T = \sqrt{5}+3 \text{ s}.$$

Question 13 (D)

Let x be the distance travelled by the skydiver. We are given that $v\dfrac{dv}{dx} = g - 0.01v^2$, where $x(v=0)=0$, and so by separation of variables,

$$\frac{dx}{dv} = \frac{v}{g-0.01v^2} \implies x(v=30) = \int_0^{30} \frac{v}{g-0.01v^2}\,dv.$$

Question 14 (C)

Using the diagram, we have that

$$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BC} + \frac{1}{2}\overrightarrow{CD} = \overrightarrow{OB} + \overrightarrow{BC} + \frac{1}{2}\left(\overrightarrow{CB} + \overrightarrow{BD}\right).$$

Since the triangles are congruent, $\overrightarrow{OA} = \overrightarrow{BC} = \underset{\sim}{a}$ and $\overrightarrow{OB} = \overrightarrow{BD} = \underset{\sim}{b}$. Hence,

$$\overrightarrow{OM} = \underset{\sim}{b} + \underset{\sim}{a} + \frac{1}{2}(\underset{\sim}{b} - \underset{\sim}{a}) = \frac{1}{2}\underset{\sim}{a} + \frac{3}{2}\underset{\sim}{b}.$$

Question 15 (C)

The angle between the vectors is

$$\theta = \arccos\left(\frac{(-\underset{\sim}{i}+\underset{\sim}{k})\cdot(2\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k})}{|-\underset{\sim}{i}+\underset{\sim}{k}|\,|2\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k}|}\right) = \frac{3\pi}{4}.$$

Question 16 (C)

First, we express the paths of the particles in Cartesian form.

Particle A: $t = y+3$ and $x = t^2 - 2t \implies x = y^2 + 4y + 3, \quad x \geq -1$ and $y \geq -3$.

Particle B: $t = y+2$ and $x = 1-t \implies x = -y-1, \quad x \leq -1$ and $y \geq -2$.

Solving these equations simultaneously gives $(x,y) = (3,-4)$ or $(x,y) = (0,-1)$. However, noting the domains and ranges of the paths, they intersect at $(0,-1)$ only.

Question 17 (B)

A point on the plane is $Q(1,0,0)$, so the distance between P and the plane is the magnitude of the scalar resolute of $\overrightarrow{PQ}$ in the direction of the plane's normal vector $\underset{\sim}{n} = \underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}$. That is, the distance is

$$d = \left|\overrightarrow{PQ} \cdot \hat{\underset{\sim}{n}}\right| = \left|2\underset{\sim}{k} \cdot \frac{1}{3}\left(\underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}\right)\right| = \frac{4}{3}.$$

Question 18 (D)

The area of the triangle ABC is

$$A = \frac{1}{2}\left|\overrightarrow{AB} \times \overrightarrow{AC}\right| = \frac{1}{2}\left|\left(2\underset{\sim}{i} + 2\underset{\sim}{j} - 2\underset{\sim}{k}\right) \times \left(8\underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}\right)\right| = 10\sqrt{2}.$$

Question 19 (A)

From the formula sheet,

$$\text{sd}(3X - 2Y) = \sqrt{3^2 \cdot 1^2 + (-2)^2 \cdot 2^2} = 5.$$

Question 20 (C)

The width of the confidence interval is two times the margin of error. That is,

$$18.8532 - 18.5468 = 2 \cdot 1.64485... \frac{2}{\sqrt{n}} \implies n = 461.$$

End of Section A

Section B

Question 1a (1 mark)

MARK 1. Provides correct answer

We have that

$$f'(x) = \frac{x^3+1}{x^2} = 0 \implies x = -1$$

and $f(-1) = \frac{3}{2}$, so the stationary point of the graph of f is located at $\left(-1, \frac{3}{2}\right)$.

Question 1b (2 marks)

MARK 1. Finds $f''(x)$, or equivalent merit
MARK 2. Provides correct answer

Differentiating f again, we have that

$$f''(x) = \frac{x^3-2}{x^3} = \frac{\left(x-\sqrt[3]{2}\right)\left(x^2+\sqrt[3]{2}x+\sqrt[3]{4}\right)}{x^3} = 0 \implies x = \sqrt[3]{2}.$$

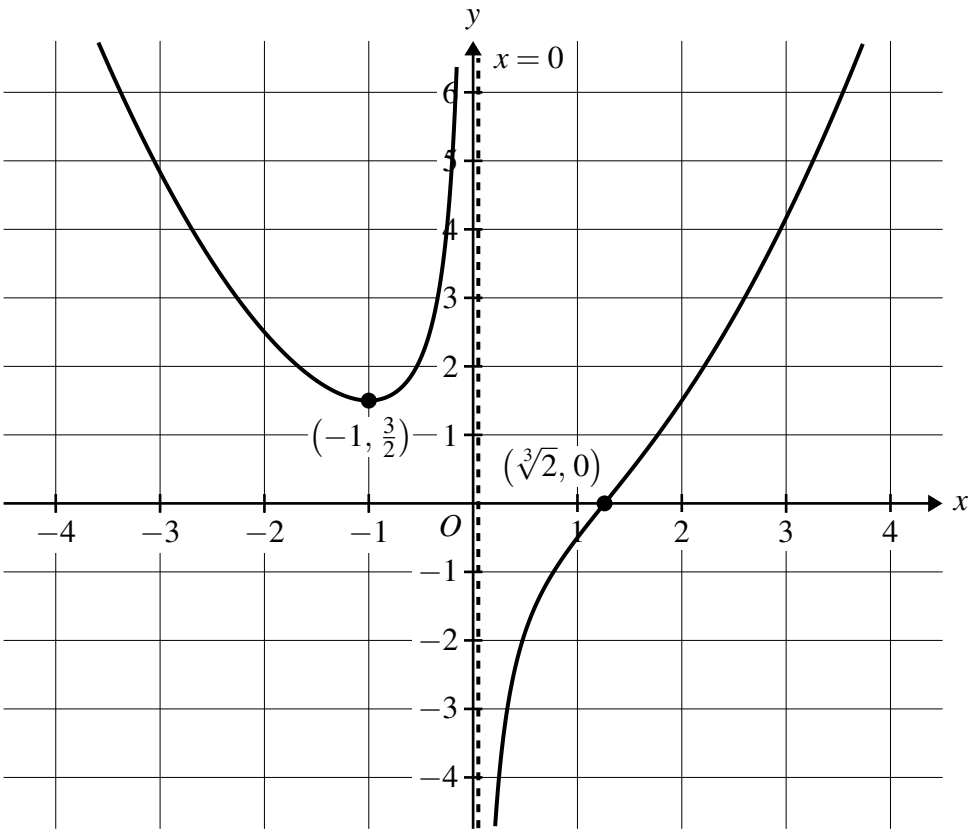
Since the concavity of f changes here and $f\left(\sqrt[3]{2}\right) = 0$, the point of inflection of the graph of f is located at $\left(\sqrt[3]{2}, 0\right)$.

Question 1c (3 marks)

MARK 1. Labels vertical asymptote

MARK 2. Labels stationary point and point of inflection

MARK 3. Shows correct graph shape



Question 1d (2 marks)

MARK 1. Expresses volume as integral

MARK 2. Provides correct answer

The volume of the solid is

$$V = \pi \int_{-2}^{-1} \left(f(x)\right)^2 dx = \frac{71\pi}{20}.$$

Question 1e (1 mark)

MARK 1. Provides correct answer

The function f has a simple zero where $x = \sqrt[3]{2}$ and a simple pole where $x = -\frac{k}{2}$ unless k is chosen so that these coincide. In this case, f has a point of discontinuity and no vertical asymptotes, and occurs when $k = -2^{\frac{4}{3}}$.

Question 2a (1 mark)

MARK 1. Provides correct answer

In polar form, $u = -1 + \sqrt{3}i = 2\,\text{cis}\left(\frac{2\pi}{3}\right)$.

Question 2b (2 marks)

MARK 1. Substitutes $z = x + yi$ into relation

MARK 2. Shows sufficient and correct algebraic work to arrive at conclusion

Where $z = x + yi$ with $x, y \in \mathbb{R}$, the relation $|z| = |z - u|$ becomes

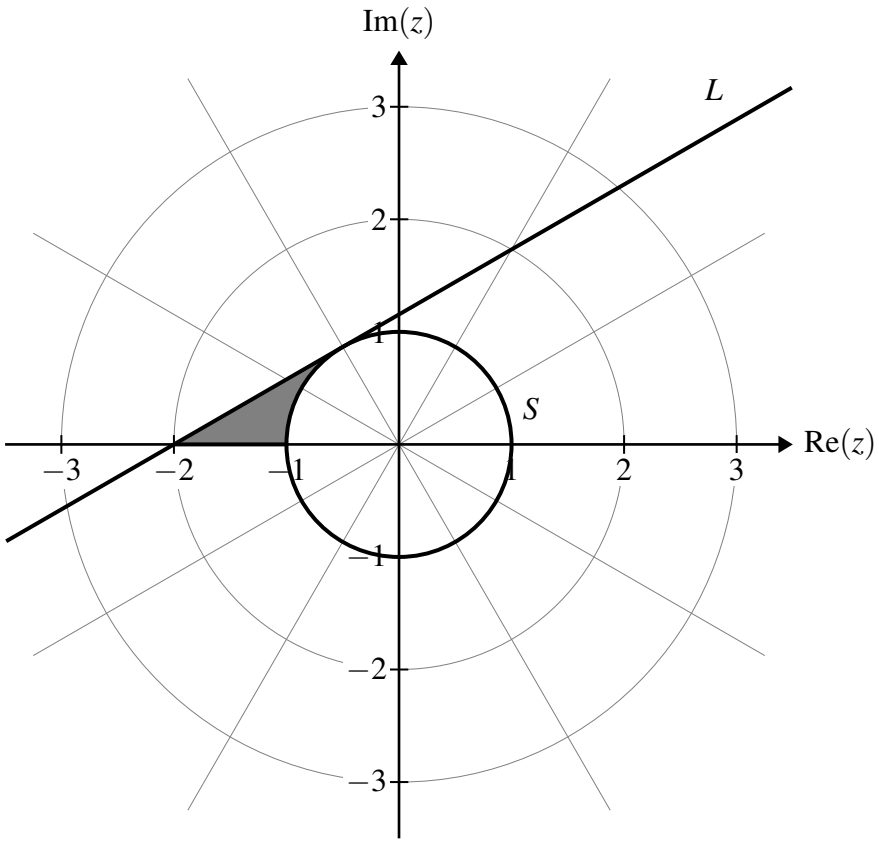
$$\begin{aligned}\sqrt{x^2+y^2} = \sqrt{(x-1)^2 + \left(y-\sqrt{3}\right)^2} &\implies x^2+y^2 = x^2 - 2x + 1 + y^2 - 2\sqrt{3}y + 3 \\ &\implies -2x - 2\sqrt{3}y + 4 = 0 \\ &\implies y = \frac{x}{\sqrt{3}} + \frac{2}{\sqrt{3}}.\end{aligned}$$

Question 2c (2 marks)

MARK 1. Sketches and labels L correctly

MARK 2. Sketches and labels S correctly

Note that the diagram below contains the solution to Question 2e.i.



Question 2d (2 marks)

MARK 1. Provides correct answer for w

MARK 2. Provides correct answer for θ

The component of the line L in the upper half plane is simply a ray originating at the point $-2+0i$ which forms an angle of $\tan\left(\frac{1}{\sqrt{3}}\right)$ with the positive direction of the real axis, measured anticlockwise. That is, $w=-2$ and $\theta=\frac{\pi}{6}$.

Question 2e.i (1 mark)

MARK 1. Shades correct region

The solution is shown on the diagram for the solution to Question 2c.

Question 2e.ii (2 marks)

MARK 1. Reasons valid method

MARK 2. Provides correct answer

The area of the shaded region can be found by subtracting the area of a sector of S from the triangle with vertices at $-2+0i$, $\operatorname{cis}\left(\frac{2\pi}{3}\right)$ and the origin. That is,

$$A=\frac{1}{2}\cdot 2\cdot\frac{\sqrt{3}}{2}-\frac{1}{2}\cdot 1^2\cdot\frac{\pi}{3}=\frac{\sqrt{3}}{2}-\frac{\pi}{6}.$$

Question 3a (2 marks)

MARK 1. Expresses $\dfrac{dW}{dt}$ in terms of $\dfrac{dw}{d\tau}$ using chain rule

MARK 2. Shows sufficient and correct algebraic work to arrive at conclusion

By the chain rule, we get

$$\frac{dW}{dt}=\frac{dW}{dw}\frac{dw}{d\tau}\frac{d\tau}{dt}=rK\frac{dw}{d\tau},$$

and so the original differential equation becomes

$$rK\frac{dw}{d\tau}=rKw\left[1-\log_e\left(\frac{Kw}{K}\right)\right]\implies\frac{dw}{d\tau}=w\big(1-\log_e(w)\big).$$

Question 3b (3 marks)

MARK 1. Applies separation of variables, or equivalent merit
MARK 2. Establishes general solution by antidifferentiation, or equivalent merit
MARK 3. Provides correct answer

By separation of variables, we have

$$\tau = \int_{e^{-1}}^{w} \frac{1}{\upsilon\big(1-\log_e(\upsilon)\big)}\,d\upsilon = \Big[-\log_e\big(\big|1-\log_e(\upsilon)\big|\big)\Big]_{\upsilon=e^{-1}}^{\upsilon=w}.$$

Because of the initial condition, the absolute value brackets can be dropped, and so

$$\tau = -\log_e\big(1-\log_e(w)\big) + \log_e\big(1-(-1)\big) = \log_e\left(\frac{2}{1-\log_e(w)}\right).$$

Rearranging for w, we get

$$1-\log_e(w) = 2e^{-\tau} \implies w(\tau) = e^{1-2e^{-\tau}}.$$

Note: the integration variable must not be used in the terminals.

Question 3c (1 mark)

MARK 1. Provides correct answer

From **part b**, $w(\tau) \to e$ as $\tau \to \infty$.

Note: alternatively, observe that $w \equiv e$ is a stable equilibrium solution to the scaled differential equation.

Question 3d.i (2 marks)

MARK 1. Applies product and chain rule
MARK 2. Shows sufficient and correct algebraic work to arrive at conclusion

Differentiating implicitly, we have

$$\frac{d^2w}{d\tau^2} = \left(1-\log_e(w)-\frac{w}{w}\right)\frac{dw}{d\tau} = -w\log_e(w)\big(1-\log_e(w)\big) = w\log_e(w)\big(\log_e(w)-1\big).$$

Question 3d.ii (1 mark)

MARK 1. Provides correct answer

The rate of tumour growth is maximised when $\dfrac{d^2w}{d\tau^2} = 0$, and since $0 < w(\tau) < e$ for all $\tau > 0$, this occurs when $\log_e(w) = 0 \implies w = 1$.

Question 3e (2 marks)

MARK 1. Reasons valid method

MARK 2. Provides correct answer

First, observe that for $w > 0$,

$$\frac{dw}{d\tau} = w\big(1 - \log_e(w)\big) - \lambda w = 0 \implies w = e^{1-\lambda}.$$

So, for all initial conditions $w(0) \in \left(0, e^{-1}\right]$, we have that $w(\tau) \to e^{1-\lambda}$ monotonically as $\tau \to \infty$. Therefore, $w(\tau) \leq e^{-1}$ for all $\tau \geq 0$ provided $e^{1-\lambda} \leq e^{-1} \implies \lambda \geq 2$.

Question 4a (2 marks)

MARK 1. Applies double angle formula, or equivalent merit

MARK 2. Shows sufficient and correct algebraic work to arrive at conclusion

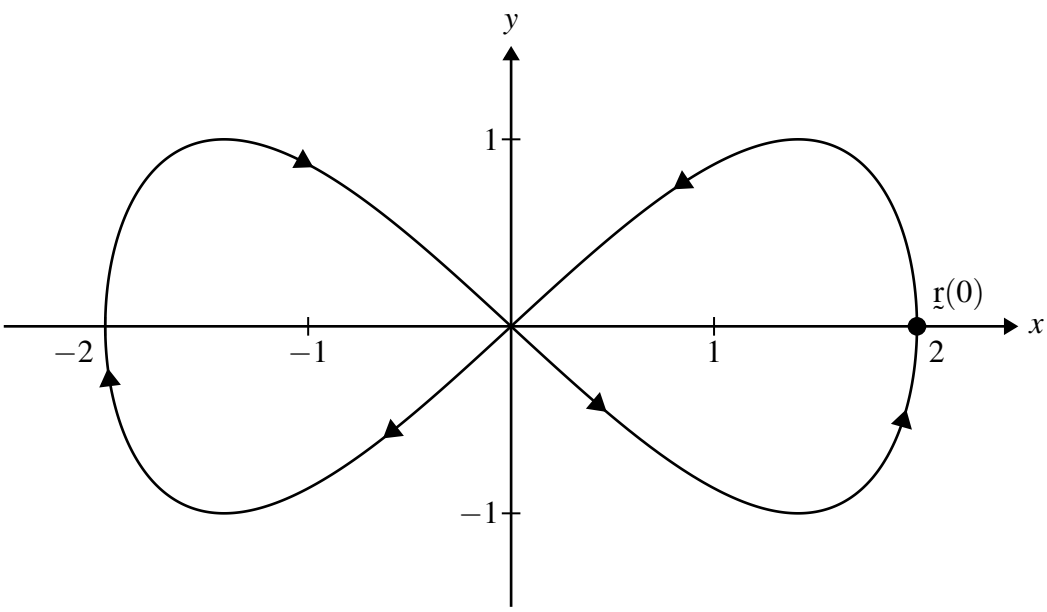
Using the double angle formula for sine and the Pythagorean identity, we get

$$y^2 = \sin^2(t) = 4\cos^2\left(\frac{t}{2}\right)\sin^2\left(\frac{t}{2}\right) = 4\cos^2\left(\frac{t}{2}\right)\left[1 - \cos^2\left(\frac{t}{2}\right)\right].$$

Substituting in $\cos^2\left(\frac{t}{2}\right) = \frac{x^2}{4}$ gives the Cartesian equation $y^2 = x^2\left(1 - \frac{x^2}{4}\right)$.

Question 4b (1 mark)

MARK 1. Labels initial position and shows direction of motion correctly



Question 4c (1 mark)

MARK 1. Provides correct answer

The time taken for the particle to first return to its initial position, i.e. the period of $\underset{\sim}{r}$, is 4π seconds.

Question 4d (2 marks)

MARK 1. Expresses total distance travelled as integral
MARK 2. Provides correct answer

The total distance travelled is

$$L = \int_0^{4\pi} \left|\underset{\sim}{r}'(t)\right| dt = \int_0^{4\pi} \sqrt{\sin^2\left(\frac{t}{2}\right) + \cos^2(t)}\, dt = 12.19 \text{ m} \quad \text{(2DP)}.$$

Question 4e (3 marks)

MARK 1. Reasons valid method
MARK 2. Provides correct answer for time
MARK 3. Provides correct answer for minimum distance

Let $d(t)$ be the distance from the particle to the point $(-1,0)$. Then,

$$d(t) = \left|\underset{\sim}{r}(t) - (-\underset{\sim}{i})\right| = \sqrt{\left(2\cos\left(\frac{t}{2}\right) + 1\right)^2 + \sin^2(t)}.$$

Then, we can solve $d'(t) = 0$ for $t \in [0, 4\pi]$ to get

$$d'(t) = 0 \implies t = 0,\ 3.69...,\ 5.13...,\ 2\pi,\ 7.44...,\ 8.88...,\ 4\pi.$$

Using the graph of d for assistance, the first time the distance is minimum is when $t = 3.69$ s (2DP), and this minimum distance is $d(3.69...) = 0.69$ m (2DP).

Question 5a (2 marks)

MARK 1. Reasons valid method
MARK 2. Provides correct answer

The x-z plane is given by $y = 0$. Therefore, the line intersects this plane where $-1 - 2t = 0 \implies t = -\frac{1}{2}$. Substituting this back into $\underset{\sim}{r}_1$, the point of intersection is $\left(\frac{1}{2},\ 0,\ 1\right)$.

Question 5b (1 mark)

MARK 1. Shows sufficient and correct algebraic work to arrive at conclusion

We let all sides of the Cartesian equations be $s \in \mathbb{R}$ so that $x = -3s$, $y = s$, and $z = 1 + 3s$. Then, a vector equation for the line L_2 is given by

$$\underset{\sim}{r}_2(s) = x\underset{\sim}{i} + y\underset{\sim}{j} + z\underset{\sim}{k} = -3s\underset{\sim}{i} + s\underset{\sim}{j} + (1+3s)\underset{\sim}{k} = \underset{\sim}{k} + s\left(-3\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}\right).$$

Question 5c (2 marks)

MARK 1. Explains why lines are not parallel
MARK 2. Shows lines do not intersect

We need to show that the lines are not parallel and do not intersect. They are not parallel since their direction vectors $\underset{\sim}{i} - 2\underset{\sim}{j}$ and $-3\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}$ are not proportional. Now, consider the system

$$\underset{\sim}{r}_1(t) = \underset{\sim}{r}_2(s) \implies \begin{cases} 1 + t = -3s \\ -1 - 2t = s \\ 1 = 1 + 3s \end{cases}.$$

The third equation gives $s = 0$. Using this, the first equation implies $t = -1$ yet the second equation implies $t = -\frac{1}{2}$, so the system has no solutions. That is, the lines do not intersect. Hence, they are skew.

Question 5d (2 marks)

MARK 1. Reasons valid method
MARK 2. Provides correct answer

The points $A(1,-1,1)$ and $B(0,0,1)$ lie on L_1 and L_2 respectively. Let

$$\underset{\sim}{n} = \left(-3\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}\right) \times \left(\underset{\sim}{i} - 2\underset{\sim}{j}\right) = 6\underset{\sim}{i} + 3\underset{\sim}{j} + 5\underset{\sim}{k}$$

be a vector perpendicular to direction vectors of both lines. The distance between the lines is then the magnitude of the scalar resolute of $\overrightarrow{AB}$ in the direction of $\underset{\sim}{n}$. That is

$$d = \left|\overrightarrow{AB} \cdot \hat{\underset{\sim}{n}}\right| = \left|\left(-\underset{\sim}{i} + \underset{\sim}{j}\right) \cdot \frac{1}{\sqrt{70}}\left(6\underset{\sim}{i} + 3\underset{\sim}{j} + 5\underset{\sim}{k}\right)\right| = \frac{3\sqrt{70}}{70}.$$

Question 5e (3 marks)

MARK 1. Finds normal vector to both planes
MARK 2. Provides correct answer for P_1
MARK 3. Provides correct answer for P_2

Since the planes contain both the direction vectors of the lines, a normal vector to both planes is $\underset{\sim}{n}$ as calculated in **part d**. Therefore, for some constants a and b, the planes are given by

$$P_1 : \ 6x + 3y + 5z = a, \quad P_2 : \ 6x + 3y + 5z = b.$$

Since P_1 and P_2 contain the points $(1,-1,1)$ and $(0,0,1)$ respectively, we have $a = 8$ and $b = 5$. Hence,

$$P_1 : \ 6x + 3y + 5z = 8, \quad P_2 : \ 6x + 3y + 5z = 5.$$

Question 6a (1 mark)

MARK 1. Provides correct answer

Suitable hypotheses are $H_0 : \mu \leq 40$ and $H_1 : \mu > 40$.

*Note: the null hypothesis should be inexact as it reflects the circumstance where a driver is **not** speeding.*

Question 6b.i (2 marks)

MARK 1. Calculates sample mean, or equivalent merit
MARK 2. Provides correct answer

Let $\overline{X} \sim N\left(\mu, \frac{4}{3}\right)$ be the random variable for the mean recorded speed. The test statistic for the van is

$$\frac{42.2+43.4+41.9}{3} = 42.5 \text{ km h}^{-1}.$$

Therefore, the p value for the test in the best case is

$$p = \Pr\left(\overline{X} > 42.5 \mid \mu = 40\right) = \Pr\left(Z > \frac{42.5-40}{\frac{2}{\sqrt{3}}}\right) = 0.0152 \quad \text{(4DP)}.$$

Question 6b.ii (2 marks)

MARK 1. Provides correct answer with justification

As $p > 0.01$, the null hypothesis should **not** be rejected at the 1% level of significance and therefore, traffic police should **not** issue a fine to the driver.

Note: the question specifically asks whether or not traffic police should issue a fine. Simply stating that the null hypothesis should not be rejected is insufficient.

Question 6c (1 mark)

MARK 1. Provides correct answer

The critical sample mean, c, is given by

$$\Pr\left(\overline{X} > c \mid \mu = 40\right) = 0.01 \implies c = 42.69 \text{ km h}^{-1} \quad \text{(2DP)}.$$

Question 6d.i (1 mark)

MARK 1. Provides correct answer

A type II error occurs when the null hypothesis is not rejected when it is false. In this context, this occurs when a fine is not issued to a speeding driver.

Question 6d.ii (2 marks)

MARK 1. Expresses type II error probability in terms of standard normal variable
MARK 2. Provides correct answer

Let v be the true speed of the vehicle. Then

$$0.1 = \Pr(\text{II} \mid \mu = v) = \Pr\left(\overline{X} < c \mid \mu = v\right) = \Pr\left(Z < \frac{c-v}{\frac{2}{\sqrt{3}}}\right).$$

Using the inverse normal cumulative distribution function, we get

$$\frac{c-v}{\frac{2}{\sqrt{3}}} = -1.281552... \implies v = 44.2 \text{ km h}^{-1} \quad \text{(1DP)}.$$

Question 6e (3 marks)

MARK 1. Expresses probability in terms of sum of remaining two measurements
MARK 2. Finds distribution of sum of remaining two measurements
MARK 3. Provides correct answer

Explicitly, let X_1, X_2 and X_3 be the independent measurements and suppose we know that $X_1 = 44.6$. Then the probability that the driver of the car is fined is given by

$$\Pr\left(\overline{X} > c \mid \mu = 43\right) = \Pr\left(\frac{X_1 + X_2 + X_3}{3} > c \,\middle|\, \mu = 43\right) = \Pr\left(X_2 + X_3 > 83.45871... \mid \mu = 43\right).$$

Now, since X_2 and X_3 are independent normal variables, $X_2 + X_3 \sim N(2\mu, 8)$. Therefore, the driver is fined with probability

$$\Pr\left(X_2 + X_3 > 83.45871... \mid \mu = 43\right) = 0.816 \quad \text{(3DP)}.$$

End of Solutions Book

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